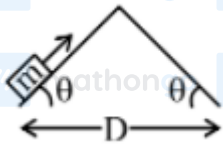
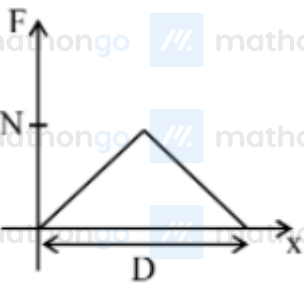


Q1 2021 (01 Sep Shift 2)

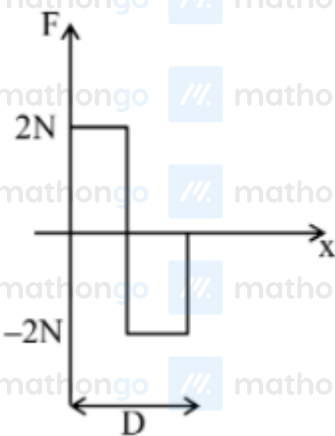
An object of mass 'm' is being moved with a constant velocity under the action of an applied force of 2 N along a frictionless surface with following surface profile.



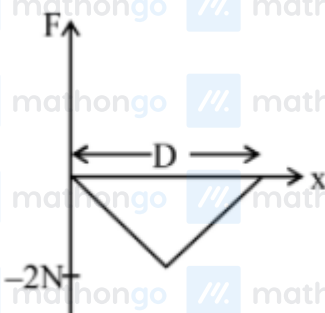
The correct applied force vs distance graph will be:



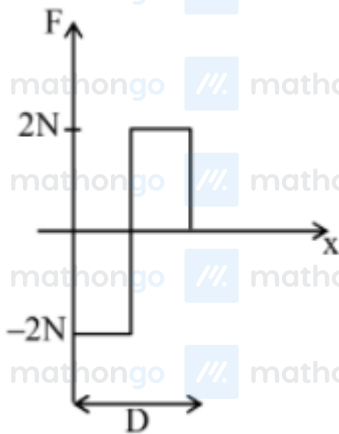
(1)



(2)



(3)

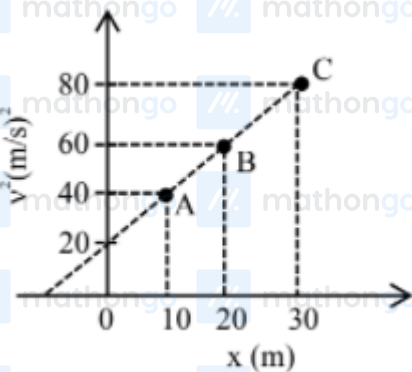


(4)

Q2 2021 (31 Aug Shift 2)

A particle is moving with constant acceleration 'a'.

Following graph shows v^2 versus x (displacement) plot. The acceleration of the particle is m/s^2 .



Q3 2021 (27 Aug Shift 2)

Water drops are falling from a nozzle of a shower onto the floor, from a height of 9.8 m. The drops fall at a regular interval of time. When the first drop strikes the floor, at that instant, the third drop begins to fall.

Locate the position of second drop from the floor when the first drop strikes the floor.

(1) 4.18 m

(2) 2.94 m

(3) 2.45 m

(4) 7.35 m

Q4 2021 (27 Aug Shift 1)

If the velocity of a body related to displacement x is given by $v = \sqrt{5000 + 24x}$ m/s, then the acceleration of the body is m/s²

Q5 2021 (26 Aug Shift 1)

Two spherical balls having equal masses with radius of 5 cm each are thrown upwards along the same vertical direction at an interval of 3 s with the same initial velocity of 35 m/s, then these balls collide at a height of

..... m. (Take $g = 10$ m/s²)

Answer Key

Q1 (2)

Q2 (1)

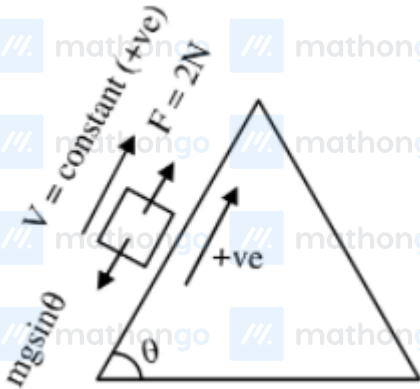
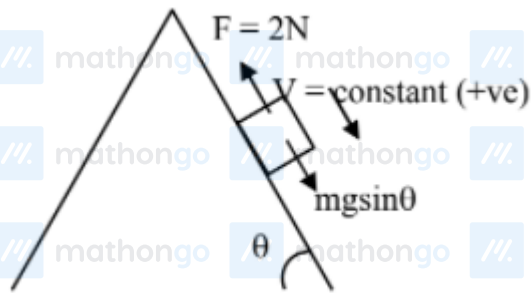
Q3 (4)

Q4 (12)

Q5 (50.0)

Q1 (2)

During upward motion


 $F = 2 \text{ N} = (+ve) \text{ constant}$ During downward motion

 $\Rightarrow F = 2 \text{ N} = (-ve) \text{ constant}$
 $\Rightarrow$ Best possible answer is option (2)

Q2 (1)

$$y = mx + C$$

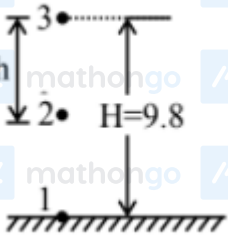
$$v^2 = \frac{20}{10}x + 20$$

$$v^2 = 2x + 20$$

$$2v \frac{dv}{dx} = 2$$

$$\therefore a = v \frac{dv}{dx} = 1$$

Q3 (4)



$$H = \frac{1}{2} g t^2$$

$$\frac{9.8 \times 2}{9.8} = t^2$$

$$t = \sqrt{2} \text{ sec}$$

Δt : time interval between drops

$$h = \frac{1}{2} g (\sqrt{2} - \Delta t)^2$$

$$0 = \frac{1}{2} g (\sqrt{2} - 2\Delta t)^2$$

$$\Delta t = \frac{1}{\sqrt{2}}$$

$$h = \frac{1}{2} g \left(\sqrt{2} - \frac{1}{\sqrt{2}} \right)^2 = \frac{1}{2} \times 9.8 \times \frac{1}{2} = \frac{9.8}{4} = 2.45 \text{ m}$$

$$H - h = 9.8 - 2.45 \\ = 7.35 \text{ m}$$

Q4 (12)

$$V = \sqrt{5000 + 24x}$$

$$\frac{dV}{dx} = \frac{1}{2\sqrt{5000 + 24x}} \times 24 = \frac{12}{\sqrt{5000 + 24x}}$$

$$\text{now } a = V \frac{dV}{dx}$$

$$= \sqrt{5000 + 24x} \times \frac{12}{\sqrt{5000 + 24x}}$$

$$a = 12 \text{ m/s}^2$$

Q5 (50.0)



When both balls will collide

$$\begin{aligned}
 y_1 &= y_2 \\
 35t - \frac{1}{2} \times 10 \times t^2 &= 35(t - 3) - \frac{1}{2} \times 10 \times (t - 3)^2 \\
 35t - \frac{1}{2} \times 10 \times t^2 &= 35t - 105 - \frac{1}{2} \times 10 \times t^2 \\
 &\quad - \frac{1}{2} \times 10 \times 3^2 + \frac{1}{2} \times 10 \times 6t \\
 0 &= 150 - 30t \\
 t &= 5 \text{ sec}
 \end{aligned}$$

$\therefore$ Height at which both balls will collide

$$\begin{aligned}
 h &= 35t - \frac{1}{2} \times 10 \times t^2 \\
 &= 35 \times 5 - \frac{1}{2} \times 10 \times 5^2 \\
 h &= 50 \text{ m}
 \end{aligned}$$

Ans. 50.00